QQ-A-200/12D
April 27, 1971
SUPERSEDING
Fed. Spec. QQ-A-200/12C
March 8, 1967

FEDERAL SPECIFICATION

ALUMINUM ALLOY BAR, ROD, SHAPES, TUBE, AND WIRE, EXTRUDED, 7079

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-200.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers specific requirements for aluminum alloy 7079 bar, rod, shapes, tube, and wire produced by extrusion. The general requirements are covered in QQ-A-200.

1.2 Classification.

1.2.1 Tempers. Bar, rod, shapes, tube, and wire shall be classified as O, T6, T62, T6510, or T6511 temper, as specified (see 6.2). The definitions of these tempers shall be as specified in ANSI Std. H35.1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

QQ-A-200 - Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube, and Wire, Extruded; General Specification for.

Federal Standard:

Fed. Std. No. 184 - Identification Marking of Aluminum, Magenesium, and Titanium.

FSC 4710, 9525,
9530, 9540

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle WA.

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Military Specification:

MIL-I-8950 - Inspection, Ultrasonic, Wrought Metals, Process for.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute ANSI Standard:

H35.1 - Alloy and Temper Designation Systems for Alumimum.

(Application for copies should be addressed to the Amaerican National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to the requirements specified in table I.

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TABLE I. Chemical composition 1/

Element	Percent Minimum	Maximum
Zinc	3.8	4.8
Magnesium	2.9	3.7
Copper	0.40	0.8
Chromium	.10	.25
Manganese	.10	.30
Iron	-	.40
Silicon	-	.30
Titanium	-	.10
Other elements, each	-	.05
Other elements, total	-	.15
Aluminum	Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in the table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties in the direction of extrusion shall conform to the requirements specified in table II. The elongation requirements shall not be applicable to the following:

(a) Material of such dimensions that a standard test specimen cannot be taken in accordance with QQ-A-200, and of such shape that it cannot be satisfactorily tested in full section.
(b) Material thinner than 0.062 inch (nominal).

3.2.2 Mechanical properties after heat-treatment. In addition to conforming to the requirement of 3.2.1, the mechanical properties of material ordered in the annealed (O) temper and subsequently properly solution heat-treated and artifically aged shall conform to the requirements of table II for the T62 temper. Material as received in the T6, T6510, and T6511 tempers shall after proper re-solution heat-treatment and artificial aging, be capable of conforming to the requirement specified in table II for the T6 and T62 tempers.

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TABLE II. Mechanical properties

Temper	Thickness, (bar and shapes); diameter, (rod and wire); wall thickness, (tube)	Cross-sectional Area	Tensile strength, minimum	Yield strength, at 0.2 percent offset, minimum	Elongation in 2 in. or 4 times D 2/, minimum
0	Inches All	Square inches All	P.s.i. 1/42,000	P.s.i. 1/24,000	Percent 10
T6, T62 3/, T6510, and T6511	Up thru 0.249	Up thru 20	75,000	67,000	7
	0.250 to 0.499	Up thru 20	77,000	68,000	7
	0.500 to 1.499	Up thru 20	78,000	70,000	7
	1.500 to 2.999	Up thru 20	79,000	70,000	7
	3.000 to 4.499	Up thru 20	79,000	70,000	7
		Over 20 thru 32	77,000	70,000	7
		Over 32 thru 50	76,000	68,000	7
	4.500 to 5.000	Up thru 38	78,000	68,000	6
		Over 38 thru 60	76,000	68,000	6
	5.001 to 5.999	Up thru 38	78,000	68,000	6
		Over 38 thru 60	76,000	68,000	6
	6.000 to 6.999	Up thru 50	76,000	66,000	6
		Over 50 thru 60	74,000	64,000	4

1/ Maximum.
2/ D represents specimen diameter.
3/ Material in the T62 temper is not available from the materials producers.

3.3 Internal defects. When specified (see 6.2), bar, rod, and shapes shall be ultrasonically inspected (see QQ-A-200). Acceptance limits shall be as specified in table III.

TABLE III. Ultrasonic discontinuity limits 1/

Thickness	Maximum weight per piece	Maximum width to thickness ratio	Discontinuity class 2/
Inches	Pounds		
0.375 to 1.499	600	10 to 1	B
1.500 and over	600	10 to 1	A

1/ Discontinuities in excess of those listed in table III may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in noncritical areas.
2/ The discontinuity class limits are defined in MIL-I-8950.

3.4 Marking. In addition to the marking required in Fed. Std. No. 184, material in the T6, T6510, and T6511 temper shall be identified by a lot number marked in at least one location on each piece.

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4. QUALITY ASSURANCE PROVISIONS
(See QQ-A-200)

4.1 Mechanical tests after heat-treatment. From the material in the annealed (O) temper, an additional number of specimens equal to those required by QQ-A-200 shall be taken and tested after solution heat-treatment and after artificial aging, to determine compliance with 3.2.2.

5. PREPARATION FOR DELIVERY
(See QQ-A-200)

6. NOTES

6.1 Intended use. This alloy is intended for use where high strength is required and where good resistance to corrosion is not important. This alloy is noted for its high transverse properties, especially, its high elongation values. The T6510 temper is intended primarily to provide extruded materials with low residual stresses and consequent minimum distortion during machining. The T6511 temper is straightened T6510 material and may or may not distort during machining.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Form and temper required (1.2.1).

(c) Dimensions required.

(d) Requirements for sizes not specifically covered (see QQ-A-200).

(e) Whether ultrasonic inspection of material is required (3.3).

(f) Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-200).

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
QQ-A-200/12D ALUMINUM ALLOY BAR, ROD, SHAPES, TUBE, AND WIRE, EXTRUDED, 7079

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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